Directed Study in Forest Management (FRST 532)

Pathways To Low-Deforestation Future In The Democratic Republic of Congo: Backcasting Analysis of Agricultural And Demographic Drivers

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Chapter 1 Introduction

1.1 Background of the study

The Congo Basin stands as one of the world's most vital ecosystems, second only to the Amazon as the largest tropical rainforest on Earth and it span across six Central African nations including the Democratic republic of Congo with an estimated 300 million hectares of forest (Carole et al., 2013; Mayaux et al., 1999). This vast biome is of global and regional importance, sequestering two-thirds of the carbon stored in Africa's live tropical vegetation (COMIFAC, 2013). The basin is a reservoir of immense biodiversity and a means of livelihood to 60 million people living within or near them, while also contributing to the food security of an additional 40 million inhabitants in regional urban centers (Nasi et al., 2011; Megevand et al., 2012).

At the heart of this essential ecological region lies the Democratic Republic of Congo. The DRC is home to nearly two-thirds of the Congo Basin's forests, representing 67% of the region's entire forest cover that accounted for 166million hectare of forest land (de wessiege et al., 2012), making it the second largest tropical forest country in the world after Brazil (Ickowitz et al., 2015). Consequently, the fate of the Congo Basin is proportionately tied to the land-use dynamics within the DRC. Despite the vast importance of this forest, it has remained under constant threat as a result of human activities which act as both direct drivers and underlying driver of the forest loss in this region(Geist and Lambin, 2001).

Addressing the complex challenge of deforestation in the DRC requires long-term strategic foresight, especially when current actions is still contributing to the problem. Tools for fostering sustainability have expanded over the last decade (Robert et al., 2002), planning for a different future necessitates a shift in approach. Back casting offers a powerful framework for this purpose. Unlike traditional forecasting, backcasting begins by envisioning a desired future state and then works backward to identify the specific steps, policies, and milestones required to achieve it (Banister & Stead, 2004). The primary aim of this research is to examine how agricultural expansion and demographic presssure function as key drivers of deforestation in the Democratic Republic of Congo and also to study utilizes future thinking tool create a framework to identify and design actionable pathways towards a low-deforestation future for the Democratic republic of Congo by the year 2050.

1.2 Problem statement

Despite its critical importance to planetary health and regional livelihoods, the Congo Basin forest ecosystem faces escalating threats. Deforestation defined as the permanent conversion of forest land or reduction in canopy cover below 10% (Kenninen et al., 2007) is rapidly accelerating, particularly in the Democratic Republic of Congo (DRC), which accounts for more than half of all regional loss (Tchatchou et al., 2015). Research indicates a worsening trajectory: forest loss in the DRC rose from 0.22% per year (2000–2005) to 0.25% (2005–2010), representing a 14% increase and a total loss of 19,759 km² in just five years (Potapov et al., 2012; COMIFAC, 2013). This acceleration is driven by specific, intensifying pressures. Primary research identifies agricultural expansion specifically shifting cultivation by smallholders as the dominant direct driver (Tyukavina et al., 2018; Curtis et al., 2018). This is inextricably linked to the underlying cause of demographic pressure (Geist & Lambin, 2001). With the DRC's population projected to increase by 100% (doubling) by 2050 (World Bank, 2018), the demand for land is set to explode. Without intervention, the annual deforestation rate is projected to rise further to 0.31%, driven by rural population growth rates of 2-3%. This business-as-usual trajectory creates a widening gap between the reality of accelerating loss and the necessity of a sustainable future, highlighting the critical need for strategic backcasting pathways to manage these drivers.

1.3 Research Aim.

The primary aim of this study is to analyse the role of agricultural expansion and population growth as the dominant drivers of deforestation in the Democratic Republic of Congo (DRC).

Building on this analyses, the research utilizes a backcasting framework to identify and design strategic pathways for achieving a low-deforestation future by 2050.

1.4 Scope of the research

This study focuses on the Democratic Republic of Congo (DRC) because it holds most of the Congo Basin's forest and has the highest deforestation rates. Successful strategies to reduce deforestation in the DRC could have a positive impact on the wider basin and provide models for other countries to follow. This research focuses on specific causes of forest loss. It analyses agricultural expansion, particularly small-scale shifting cultivation, as a direct driver. It also examines Demographic pressure as an underlying driver. The study uses a desk-based methodology that relies on secondary data to inform a back casting analysis.

Chapter 2 literature Review

2.1 Brief overview of the DRC Forest.

The Democratic Republic of Congo (DRC) contains approximately 155 million hectares of forest (DRC, 2011), making it the largest forest area in Africa and the fifth largest in the world. Forests cover roughly 67% of the national territory (de Wasseige et al., 2009). This vast ecosystem comprises diverse types, including humid dense forest, mountain forest, transition forest, edaphic forest, mangrove forest, mosaic savannah, and secondary forest (de Wasseige et al., 2012). The moist forest alone occupies nearly 99 million hectares, of which 83 million are found at low altitude. It is estimated that 60 million hectares of this forest are suitable for timber production—an area roughly equivalent to the size of all other Central African countries combined (Debroux et al., 2007).

Hosting the largest intact tract of tropical forest in Africa, the DRC is a reservoir of immense biodiversity and holds global importance as a carbon sink for mitigating climate change, while simultaneously supporting the livelihoods of millions of people (Molua, 2019).

2.1.1 Deforestation Concepts and Trends

Deforestation is defined as the loss of forest cover or its conversion to alternative land uses, often marked by a reduction of canopy cover below a 10% threshold (Vásquez-Grandón et al., 2018; FAO, 2000a). The loss of natural forest is a critical global issue with resonant impacts ranging from significant greenhouse gas emissions (Simula & Mansur, 2011) and biodiversity

loss (Foley et al., 2007) to a reduction in water regulation (Lele, 2009). Ultimately, deforestation diminishes the forest's ability to provide the ecosystem goods and services that sustain local livelihoods (Lambin & Meyfroidt, 2011). The causes of forest disturbance are driven by multiple

synergistic factors acting together, rather than single variables alone (Geist and Lambin, 2002; Megevand, 2013), meaning that policies and responses need to address a variety of factors and

their interactions. While national rates of deforestation in the DRC are relatively low compared to countries like Brazil, they have accelerated concerningly in the last decade, nearly doubling to approximately 0.5% per year (FAO, 2020).

2.2 Theoretical Drivers

2.2.1 Agricultural Expansion

Kissinger et al. (2012) demonstrated that smallholder agriculture remains a primary cause of deforestation in tropical areas, contributing to 35% of total deforestation in Africa, 65% in Latin America, and over 30% in Southeast Asia. However, the narrative that slash-and-burn agriculture is responsible for irreversible deforestation is increasingly being challenged. Many scholars emphasize the benign character and non-disruptive ecological effects of traditional agriculture when carried out by indigenous peoples. Shifting cultivation, the dominant form of agriculture in the DRC, involves clearing land, cultivating it, and then leaving it to fallow in a cyclical fashion (Thrupp et al., 1997).

This agricultural strategy is viewed as problematic for two distinct reasons. First, the initial clearing of forests is technically classified as 'deforestation.' However, Finegan and Nasi (2004) and Makana and Thomas (2006) argue that if cleared patches are small and fallow periods are sufficiently long, fields tend to regenerate into forest and that while there may be changes in species composition, biodiversity losses in these systems are lower than in almost any other anthropogenic system. This sustainable cycle typically functions well in areas with abundant land and relatively low population density.

The second critique posits that increasing population pressure shortens fallow periods, preventing soil regeneration and leading to permanent deforestation (Mertz, 2002; Ickowitz, 2006). This is particularly evident in peri-urban areas. However, Ickowitz et al. (2015) argue it is misleading to blame shifting cultivation *per se*; rather, the driver is the high population density surrounding urban centers.

Furthermore, the proposed solution of agricultural intensification is not straightforward. While it is assumed that higher yields will spare forests, Phelps et al. (2013) and Hourticq et al. (2013) utilized models to show that intensification in the DRC can actually *increase* deforestation. This occurs because productivity growth increases the profitability of agriculture, incentivizing farmers to clear more land. Additionally, intensification efforts in sub-Saharan Africa often fail because they are labor-intensive (Mafongoya, 2006; Long et al., 2009). In the Congo Basin, labor is a binding constraint, particularly for women, who play an essential role in agriculture.

Finally, regarding future trends, several studies predict that population growth will result in the near-total loss of the DRC's primary forest by 2050 (Ernst et al., 2013; Tollens, 2010). This prediction relies on a simulation model by Zhang et al. (2002), which assumes shifting cultivators use a 2-year cropping cycle followed by a 15-year fallow. Ickowitz et al. (2015) critique this, noting that historical records show fallow periods in the DRC were rarely that long, even in 1907 when population density was much lower. Therefore, models predicting total forest loss based on rigid fallow assumptions may be overly pessimistic, though the link between subsistence farming and human pressure for food security remains a critical challenge (Mayaux et al., 2013).

2.2.2 Population Growth and Demographics

Demographic factors including population size, structure, and density act as powerful underlying drivers of deforestation in the Congo Basin (Tchatchou et al., 2015). In the DRC specifically, these factors are not merely background context but act as the primary engine intensifying pressure on natural resources.

2.2.2.1 Rapid Growth and Future Projections:

The DRC is experiencing a demographic explosion. With a high annual population growth rate of approximately 3.2% (World Bank, 2023), the national population is projected to nearly double by 2050, exceeding 200 million people (World Bank, 2018; UNDESA, 2022). This shift is driven by a decline in mortality rates combined with persistently high fertility, creating a "youth bulge" that significantly increases the demand for resources.

2.2.2.2 Mechanisms of Deforestation

Demographic pressure drives forest loss through two primary mechanisms:

- Agricultural Expansion: As the rural population expands, the demand for subsistence food production rises. Without agricultural intensification, this necessitates the conversion of more forest land into farmland, creating a direct correlation between population density and deforestation rates (Tyukavina et al., 2018).
- Energy Demand: The growing population drives an exponential demand for fuelwood and charcoal, the primary energy sources for the majority of households. This creates "halos" of forest degradation around expanding settlements (Charlotte et al., 2023).

2.2.2.3 Migration and Urbanization

While natural population growth is the primary driver, migration patterns shape where deforestation occurs. Rural-urban migration concentrates the demand for food and fuel in peri-urban areas. Spatial analysis confirms this trend, highlighting a significant correlation between population density and forest fragmentation, with deforestation "hot spots" typically located on the periphery of major cities and densely populated areas (FAO & UCL, 2013). Additionally, sudden population movements due to conflict or the emergence of extractive industries can trigger rapid, localized deforestation events (Geist et al., 2002). Thus, in the DRC context, population growth acts as a multiplier effect, exacerbating the impact of subsistence agriculture and energy extraction on the forest frontier

2.3 Theoretical Framework: Backcasting

To address the complex, long-term challenge of deforestation in the DRC, this study employs a backcasting approach. Backcasting is a normative methodological framework that begins by defining a desirable future vision in this case, a Net-Zero Deforestation economy and works backward to identify the sequence of short-, medium-, and long-term actions required to connect that future to the present (Van Bers et al., 2016).

This approach is fundamentally different from forecasting. While forecasting projects current trends forward to predict a likely future which for the DRC implies unsustainable forest loss driven by population growth, backcasting starts from the solution. It asks: "If we want to achieve this low-deforestation future, what specific policy and structural changes must occur today?" This makes it particularly suited for sustainability problems where "business-as-usual" trends are the problem itself.

Modern backcasting approaches also account for non-linearity and disruption (Ballard, 2019). This study acknowledges concepts such as "hinge-points" (Mandy et al., 2021) or "triggers"

(Haasnoot et al., 2013), which are moments where existing actions become inadequate or new opportunities arise. Furthermore, it draws on the "seeds-based pathways" concept (Bennett et al., 2016) To address the complex, long-term challenge of deforestation in the DRC, this study employs a backcasting approach. Backcasting is a normative methodological framework that begins by defining a desirable future vision in this case, a Net-Zero Deforestation economy and works backward to identify the sequence of short-, medium-, and long-term actions required to connect that future to the present (Van Bers et al., 2016).by identifying existing initiatives that are working in respect to deforestation in the region and the strategy and the pathway they can curb deforestation with respect to demographic and agricultural expansion in the area.

Chapter 3 Methodology

3.1 Research Design

This study employs a qualitative, desk-based research design grounded in a normative backcasting framework. The research adopts a dual approach: a descriptive phase to analyse current deforestation trends and drivers, and a normative phase to define a sustainable 2050 vision and identify the strategic pathways required to achieve it.

3.2 Data Collection and Sources

To ensure reliability and comprehensive coverage, data was curated from four distinct categories of secondary sources. This tiered approach allows for both broad contextual understanding and precise quantitative analysis.

- Academic & Contextual Literature (The Foundation): Peer-reviewed journals (e.g., Geist & Lambin, Tyukavina et al.) and foundational reports (FAO, COMIFAC) were utilized to establish the theoretical understanding of deforestation drivers and identify the historical consensus regarding the role of subsistence agriculture and population dynamics.
- Spatial Data (The Diagnostic): Data from Global Forest Watch (GFW) was selected for the analytical phase to empirically quantify the extent of forest loss for the period 2001–2024. This provided up-to-date, quantitative statistics on the dominance of agricultural expansion as a primary driver.

- Demographic and Economic Scenarios (The Projections): Projections from the International Futures (IFs) forecasting model, as presented by the Institute for Security Studies (ISS) (Yeboua et al., 2021), were used to establish the baseline population trajectory and model the potential impacts of future strategic interventions.
- Policy Frameworks (The Feasibility): Key documents, including the National REDD+ Strategy and the Paris Agreement (NDCs), were analyzed to assess the political feasibility of the proposed pathways and identify existing funding mechanisms.

3.3 Analytical Framework

The backcasting analysis was conducted in four sequential steps:

- Step 1: Defining the 2050 Normative Vision: A desired future state of Net-Zero Deforestation was defined, characterized by stabilized demographics and a decoupled agricultural system.
- Step 2: Identifying the Business-as-Usual (BAU) Gap: Current trends and projections were analyzed to quantify the widening gap between the high-deforestation trajectory and the sustainable vision.
- Step 3: Developing Strategic Pathways: Two parallel pathways (Agricultural Transformation and Demographic Management) were designed, with specific actions mapped onto short-, medium-, and long-term timelines to bridge the identified gap.
- Step 4: Policy Integration: The proposed pathways were validated against existing national and international policy frameworks to ensure alignment with the DRC's development goals and climate commitments.

Chapter 4

4.1 The Vision 2050

By 2050, the Democratic Republic of Congo has transitioned to a "Net-Zero Deforestation" agricultural economy. In this vision, the agricultural system meets national food and income needs without expanding into primary forests, achieved through a 58.8% increase in crop yields and the widespread adoption of sustainable intensification (Yeboua et al., 2021).

Concurrently, population pressures on land and forests have been stabilized. By mitigating socio-economic drivers through education and health investments, the country has achieved a demographic transition to a fertility rate of approximately 2.6, reducing the exponential demand for new arable land (Yeboua et al., 2021).

Ultimately, the combined transformation of agriculture and demographic dynamics establishes a sustainable trajectory where forest conservation coexists with improved rural welfare, food security, and economic development, fulfilling the DRC's commitments under the Paris Agreement and REDD+ Strategy.

4.2 Business-as-Usual (BAU) Scenario: The Widening Gap

Under a Business-as-Usual (BAU) trajectory, the Democratic Republic of Congo (DRC) faces accelerating forest loss driven primarily by agricultural expansion and rapid population growth.

4.2.1Historical Trends (2001–2024)

Historical data reveals a marked escalation in forest loss. Figure 4.1 illustrates the trend of primary forest cover loss over the last two decades. Between 2001 and 2012, the DRC lost approximately 1.88 million hectares of primary forest. In the subsequent period (2013–2024), this figure rose sharply to 5.66 million hectares, representing a threefold increase compared to the preceding decade. Annual rates fluctuated but trended upward, peaking at 589,812 hectares in 2024. Cumulatively, this period saw the loss of approximately 7.5 million hectares of primary forest.

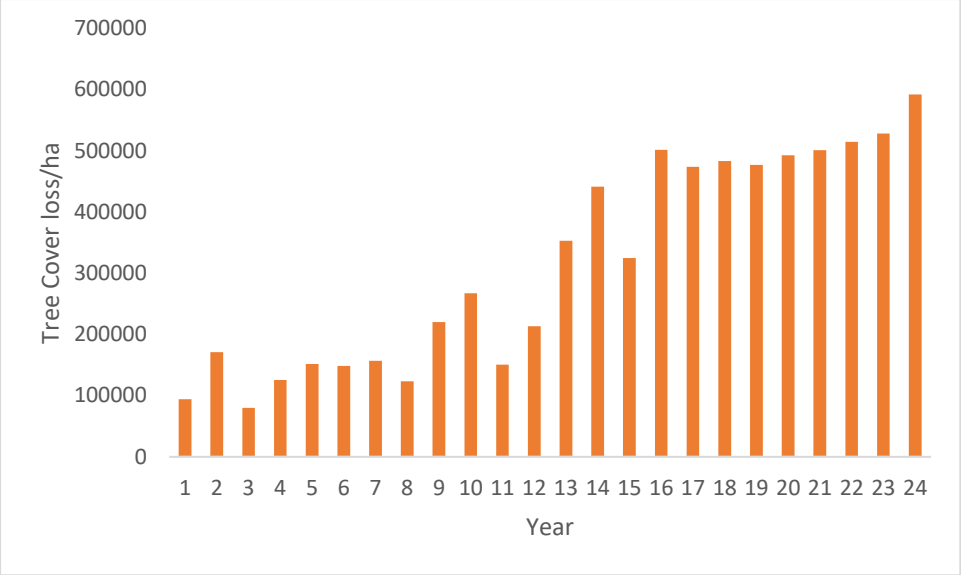

Figure 1 Annual Tree Cover Loss in the Democratic Republic of Congo (2001–2024)

Source: Author's analysis based on spatial data from Global Forest Watch (2024).

4.2.2 The Dominant Driver

The persistence of this loss is driven by specific activities. As shown in Figure 4.2, agricultural expansion is the overwhelming primary cause. Shifting cultivation accounts for 82% of total loss, while permanent agriculture contributes another 15%. Combined, agricultural activities are responsible for approximately 97% of all forest cover loss in the region.

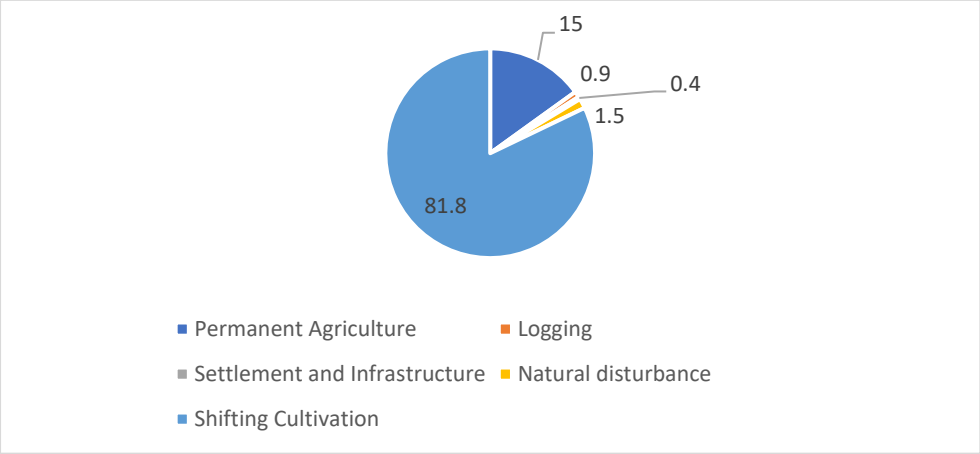

Figure 2 Proximate Drivers of Tree Cover Loss in the Democratic Republic of Congo (2001–2024)

Source: Author's analysis based on data from Global Forest Watch (2024)

4.2.3 Future Projections (2025–2050)

Looking ahead, demographic trends will amplify these pressures. National projections indicate that the DRC's population will nearly double to approximately 217 million by 2050 (fig 4.3).

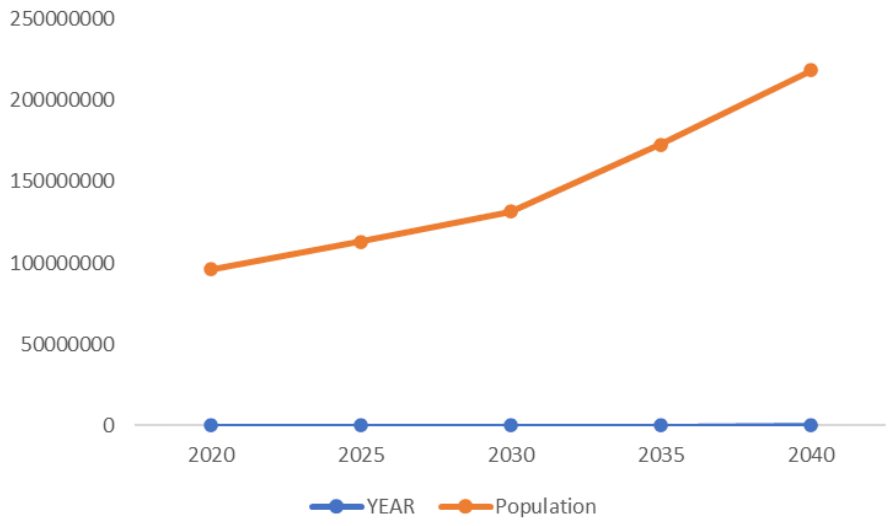

Figure 3 Population Growth Trend and Projections for the Democratic Republic of Congo (2000–2050)

Figure 4.3 presents the International Futures (IFs) model projection for deforestation rates under this scenario. The data indicates a clear upward trajectory, with the annual deforestation rate rising from 0.11 million hectares in 2020 to a projected high of 0.25 million hectares per year by 2050. Cumulatively, this trajectory results in a total projected forest loss of 5.9 million hectares between 2025 and 2050.

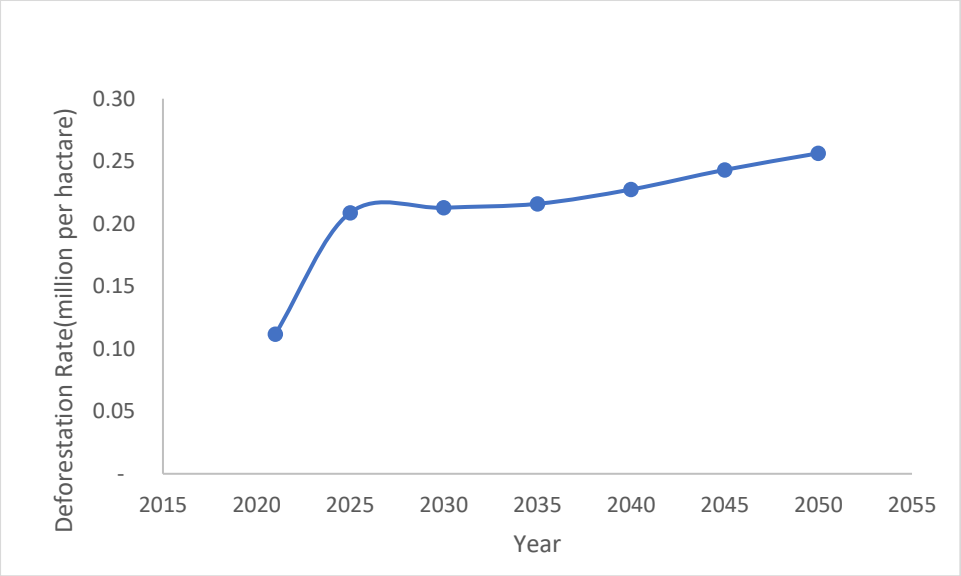

Figure 4 Projected Annual Deforestation Rates in the DRC under a Business-as-Usual Scenario (2020–2050)

Source: Author's calculation based on data from the International Futures (IFs) modeling platform (2024).

4.2.4 Consequences and the Gap

The implications of this trajectory are severe. Under current trends, the land-use sector risks shifting from a net carbon sink to a net emitter. The World Bank Group–CCDR (2023) estimates that continued loss could cost the economy US$95.3 billion in lost carbon sequestration capacity. Taken together, these trends create a widening gap between the projected high-deforestation future and the desired 2050 vision of a Net-Zero Deforestation. This BAU scenario defines the critical problem space that the following backcasting pathways aim to transform.

4.3 Strategic pathways to 2050

4.3.1 Pathway 1: Transforming Agricultural Practices

This pathway addresses the primary driver of deforestation by shifting the agricultural sector from "extensification" (clearing more land) to "intensification" (growing more on existing land). Currently, 70% of the workforce is employed in agriculture, yet the country cannot meet its food needs, driving the continuous expansion of subsistence farming into forests.

Short-Term Actions (2025–2030): Foundations & Land Security

In the short term, the priority is to establish the necessary foundations for sedentary farming. The government must engage in broad, consensual land reforms to address recurrent land disputes and protect the rights of vulnerable farmers. Secure tenure acts as the primary incentive for farmers to invest in long-term soil fertility rather than engaging in shifting cultivation. Simultaneously, the government and development partners should work hand-in-hand to generate and disseminate improved seeds and farming practices adapted to climate change. Initial investments must be directed specifically toward subsistence and smallholder farming to boost immediate yields per hectare.

Medium-Term Actions (2030–2040): Infrastructure & Value Chains

Over the medium term, the focus shifts to building the physical infrastructure required to sustain high-yield agriculture. This involves rehabilitating rural transport networks to allow farmers to sell produce easily and receive support from agricultural extension services. Improved market access makes sedentary farming profitable, reducing the economic pressure

to clear forest for survival. Additionally, adequate harvesting, storage, and marketing mechanisms must be established to reduce agricultural losses along the value chain from producer to consumer. Efforts should also focus on diversifying the DRC's productive and export bases to achieve structural economic transformation, ensuring production is primarily intended for domestic consumption.

Long-Term Outcomes (2040–2050): Decoupling & Food Security

By 2050, these interventions are projected to decouple food security from deforestation. Crop yields are expected to increase by 58.8% between 2024 and 2050, with total agricultural production reaching 91.3 million metric tons approximately 36% higher than the current path forecast. Consequently, agricultural import dependence is projected to decline to 50.3%, significantly improving national food sovereignty. The social impact of this transformation is profound: malnutrition is expected to decline by 50% (affecting 6.8 million fewer people), and 8.3 million fewer people will be living in extreme poverty. Ultimately, agricultural growth is achieved through intensification rather than extensification, effectively securing food supplies without the need for further deforestation.

4.3.2 Pathway 2: Managing Demographic Pressures

Population growth is the primary underlying driver fuelling the expansion of agricultural land. Currently, the correlation between population growth and deforestation in the DRC is high because the majority of the growing rural population relies on extensive slash-and-burn agriculture for food security. Managing demographic pressures does not imply restricting growth directly but rather addressing the socio-economic factors that intensify its environmental impacts. This pathway is grounded in the understanding that even if DR Congo manages to keep a greater part of the natural resources wealth, the prospects for human development would remain constrained by the persistently high fertility rate(World Bank, 2018).

This pathway envisions a future where DRC policymakers embrace family planning and human capital development as necessary for forest conservation. It emphasizes a comprehensive approach that targets improvements in health, education, and energy, thereby reducing the subsistence demand for new farmland.

Short-term actions (2025–2030):

The initial phase focuses on foundational changes. This includes public information campaigns aimed at raising awareness of family planning, particularly in rural areas where fertility rates are high. Slowing the rate of rural population growth is the most effective long-term strategy to stabilize the demand for agricultural land. Concurrently, efforts to improve the health system—targeting reductions in child mortality and malaria—are essential, as lower mortality rates are a prerequisite for the demographic transition to smaller family sizes. Furthermore, introducing clean cooking technologies reduces dependence on fuelwood and charcoal, breaking the cycle where population density leads to forest degradation around urban centers.

Medium-term actions (2030–2040):

The focus shifts to scaling these models. The government continues to roll out efforts to improve school enrolment and completion, particularly for girls. This phase emphasizes vocational and technical training in key growth sectors. By equipping the youth with skills for the manufacturing and service sectors, the economy creates "non-farm" livelihoods. This draws labor away from subsistence agriculture, directly reducing the number of new households entering the shifting cultivation cycle. Policies also promote balanced rural–urban development to manage migration pressures on forest-dependent regions.

Long-term actions (2040–2050):

By this stage, these interventions lead to a significant demographic transition. In this scenario, the fertility rate declines to 2.6 births per woman by 2050 (down from a projected 3.4) (Yeboua et al., 2021). This achievement improves the dependency ratio and increases the size of the economy by roughly US$15.4 billion. Crucially, lifting 15.4 million people out of extreme poverty severs the link between survival and deforestation. Wealthier, better-educated households are less dependent on shifting cultivation and more able to afford clean energy, resulting in a stabilization of the forest frontier.

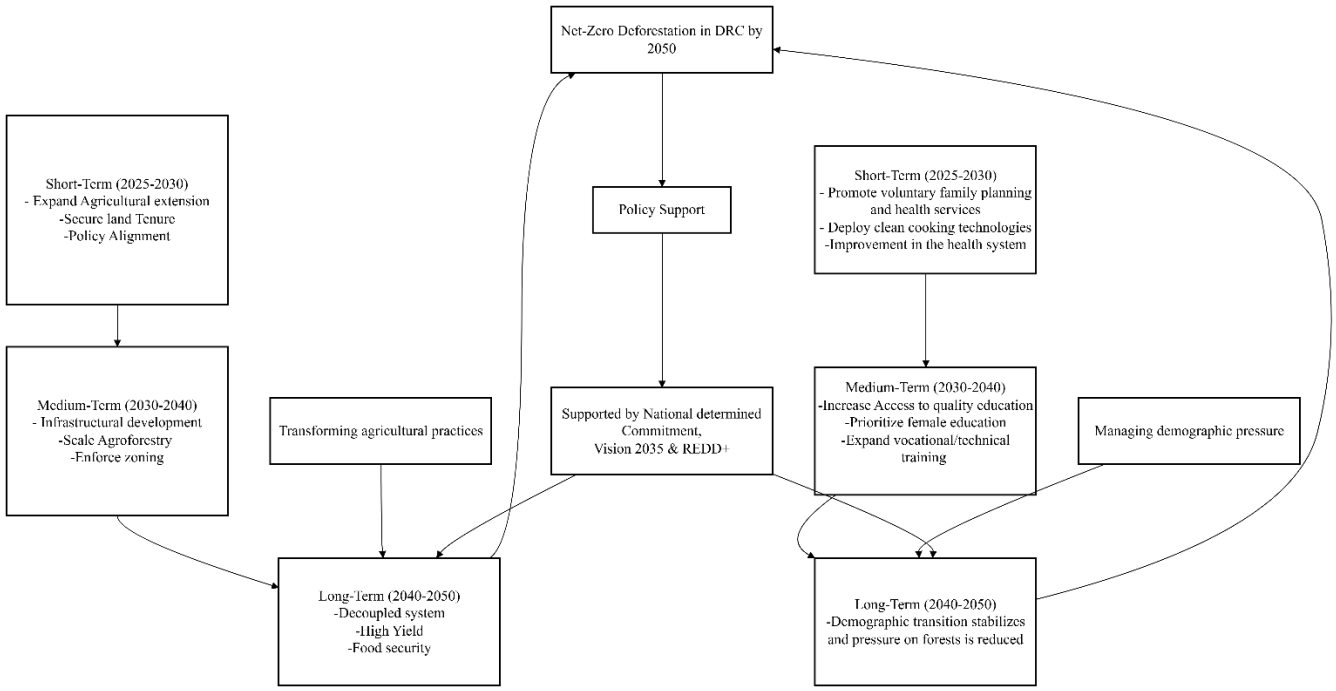

Figure 5 Strategic Backcasting Pathways for a Net-Zero Deforestation Future in the DRC (2025–2050)

4.4 Policy Alignment

The DRC has been actively engaged in REDD+ "Readiness" activities since 2009, establishing a robust institutional framework led by the Ministry of the Environment, Nature Conservation, and Tourism (MECNT), which is supported by multilateral initiatives like the FCPF and UN-REDD This process laid the groundwork for the National REDD+ Framework Strategy adopted in 2012. This strategy aims to stabilize forest cover at 63.5% by 2030, providing the overarching governance structure for the pathways proposed in this study (Aquino and Guay, 2013).

The financial feasibility of these pathways is secured through the Central African Forest Initiative (CAFI). Building on a partnership formed in 2015, a renewed 10-year agreement (2021–2031) was signed at COP 26, unlocking US$500 million to halt deforestation through sustainable & inclusive rural development (CAFI, 2025). This funding mechanism is a critical enabler, directly supporting the agricultural intensification and land tenure reforms outlined in Pathway 1, as well as the community livelihood investments in Pathway 2.

Furthermore, these environmental goals are inextricably linked to national development. The DRC's National Strategic Plan for Development (PNSD) prioritizes economic growth, employment promotion, and access to basic social services. By targeting human capital development and climate resilience by 2035 (Tchatchou et al., 2015), this national strategy

provides the political mandate for the health and education interventions proposed in Pathway 2, while simultaneously supporting the food security goals of Pathway 1. Consequently, the backcasting pathways do not require new, isolated policies, but rather the integrated implementation of these existing international and national commitments.

4.6 Critical Implementation Bottlenecks

While the proposed pathways are technically feasible, their implementation faces significant structural barriers inherent to the DRC's political economy

4.6.1 Governance and Land Tenure Complexity

The most profound barrier is the weakness of forest governance and the systemic fragility of the state. As noted by Karsenty and Ongolo (2011), the DRC functions as an archetype of a "fragile state," where a fragmentation of power allows elites to prioritize private interests over the public good. This fragility makes long-term agreements such as budget disbursements for health services or land tenure reforms unreliable and prone to disruption and this has a huge effect on the activities or implementation of REDD+ in the region.

Specifically, regarding Pathway 1, the transition to secure land tenure faces deep-seated challenges. Rakundo and Betge (2020) highlight that formalizing land rights risks exacerbating conflicts due to persisting tensions between statutory and customary systems and also there is a severe risk of "elite capture," where increased land value from agricultural intensification leads to speculation and the dispossession of vulnerable smallholders by powerful local actors. This abuse of power remains a pervasive risk that could undermine the equity of community titling schemes.

4.6.2 The Infrastructure Paradox

Pathway 1 also relies on connecting farmers to markets to make sedentary agriculture profitable. However, infrastructure development presents a critical dilemma. While improving the DRC’s dilapidated road network is imperative for reducing poverty and enabling access to agricultural inputs (Buys et al., 2006), better roads reduce the "passive protection" that isolation has historically provided forests. Increased accessibility makes the region attractive to global investors in commercial agriculture, potentially accelerating deforestation rather than reducing it unless strictly managed.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1 Discussion and Summary

First, the findings underscore that deforestation in the DRC is driven less by industrial extraction and more by subsistence survival strategies. Analysis of spatial data from 2001 to 2024 reveals that agricultural expansion accounts for 96.8% of all tree cover loss. This empirical evidence aligns with the broader literature, which frames deforestation in the Congo Basin as fundamentally a food security issue rather than one solvable solely through logging bans (Greenfield, 2024). The overwhelming predominance of shifting cultivation signals a systemic lack of agricultural intensification, pointing to the urgent need for interventions that strengthen food systems rather than merely restricting access (Villa et al., 2021).

This structural driver is intensified by a "demographic multiplier effect," whereby rapid population growth acts as the primary engine of forest loss. With the national population projected to nearly double to approximately 217 million by 2050, the demand for arable land and wood fuel is set to exceed the forest's regenerative capacity. This projection is consistent with earlier warnings by Tchatchou et al. (2015) regarding the pressure of rural population density. With rural households requiring an estimated 0.25 hectares of land each for subsistence (Tollens, 2010), this population explosion creates an unsustainable trajectory where the annual deforestation rate could rise to 0.31% by 2035 (Kibambe & Defourny, 2010). The economic stakes of this trajectory are severe, with an estimated cost of US$95.3 billion in lost carbon sequestration capacity (World Bank Group, 2023).

Consequently, the study emphasizes that a "Net-Zero Deforestation" future is achievable only through decoupling human development from resource depletion. For the agricultural sector, this means decoupling food security from land expansion through a 58.8% increase in crop yields, achieved via secure land tenure and agroforestry. This strategic focus is strongly corroborated by scientific consensus; Shapiro et al. (2021) identify climate-smart agriculture as a critical lever for reducing forest pressure, while Mangaza et al. (2021) argue for productive farming systems driven by local communities. Furthermore, the need for institutional synergy between forestry and agriculture is reinforced by Sonwa et al. (2016). In parallel, human development must be decoupled from resource depletion by targeting a demographic transition to a fertility rate of 2.6 births per woman. Supported by Yixu and Zhao (2023), who emphasize human resource development in natural resource management, these findings suggest that

systemic transformation requires integrated strategies addressing ecological sustainability, agricultural productivity, and demographic change simultaneously.

5.2 Conclusion

Deforestation in the Democratic Republic of Congo is accelerating despite over a decade of REDD+ interventions, driven overwhelmingly by shifting cultivation which accounts for 82% of tree cover loss. This persistent rise highlights a critical reality: shifting cultivation is not merely an environmental management issue but a survival strategy practiced primarily for sustenance. With the national population projected to nearly double to approximately 217 million by 2050, the pressure to clear forests for food and fuel will inevitably exceed the ecosystem's natural regenerative capacity under a 'Business-as-Usual' scenario. Consequently, achieving Net-Zero Deforestation requires a decisive step away from current approaches to fundamentally decouple human survival from forest conversion. This necessitates a structural transformation focused on agricultural intensification to ensure food security on existing lands, alongside demographic stabilization and clean energy transitions, as critical prerequisites for long-term forest sustainability.

5.3 Strategic Policy Recommendations

To bridge the gap between the BAU trajectory and the 2050 vision, this study proposes four strategic recommendations for the Government of the DRC and its international partners.

- Prioritize land tenure reform as a conservation prerequisite to accelerate the implementation of local community forest concessions
- Shift investment to agricultural intensification through direct national subsidies and REDD+ funding toward inputs for intensification (improved seeds, fertilizers) and extension services
- Integrate demographics into climate action by adopting the population, health, and environment (PHE) approach in all forest-adjacent provinces
- Operationalize land-use zoning by enforcement of zoning laws to restrict agricultural expansion

5.4 Limitations and Future Research

This study acknowledges the inherent limitations of relying on secondary data. The analysis is dependent on the accuracy of underlying demographic models and the remote sensing classification algorithms used by Global Forest Watch to identify drivers. Furthermore, as a

desk-based study, it cannot capture the real-time, hyper-local nuances of community resistance or specific local governance failures that primary fieldwork might reveal. However, the use of high-level aggregate data is appropriate and necessary for designing national-scale strategic pathways over a multi-decade horizon.

Future research should therefore prioritize field-based evaluations to ground-truth these findings. Specifically, studies should investigate community perceptions regarding the adoption of agroforestry and assess local stakeholder commitment to the proposed interventions. Such primary research is critical to understanding the practical, on-the-ground barriers facing deforestation control in the DRC that macro-level analysis may overlook.

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